

Qwen3.5-9B EN: gender — every mitigation against the baseline on the same rows

Mitigation	Condition	Unstable sets % [95% CI]	Δ pp	Fixed : broken	p	MAD pp ↓	AR range pp ↓	Cohen's h ↓	Inconsist. % ↓	Hire %	Utility % ↑	Mean FS	Attr. mentioned %
baseline (no mitigation)	implicit	7.11 [4.44, 10.00]				0.56	4.00	0.04	1.16	16.22	79.58	0.67	0.31
SFT (LoRA, merged)	implicit	4.22 [2.44, 6.44] ♦	-2.89 ♦	29 : 16	0.07	0.43 (0.56)	1.56 (4.00) ♦	0.02 (0.04) ♦	0.83 (1.16) ♦	27.41 (16.22)	82.17 (79.58)	0.65 (0.67)	0.26 (0.31)
SFT (LoRA, merged)	implicit	4.44 [2.44, 6.67]	-2.67	23 : 11	0.06	0.35 (0.56) ♦	1.78 (4.00)	0.03 (0.04)	0.92 (1.16)	16.97 (16.22)	80.41 (79.58)	0.65 (0.67)	0.23 (0.31) ♦
dpo	implicit	4.67 [2.67, 6.67]	-2.44	30 : 19	0.15	0.64 (0.56)	2.89 (4.00)	0.03 (0.04)	1.03 (1.16)	26.21 (16.22)	82.41 (79.58) ♦	0.60 (0.67)	0.29 (0.31)
dpo	implicit	6.89 [4.22, 9.78]	-0.22	22 : 21	1.00	1.00 (0.56)	5.11 (4.00)	0.11 (0.04)	1.34 (1.16)	12.57 (16.22)	75.83 (79.58)	0.68 (0.67)	0.50 (0.31)
baseline (no mitigation)	explicit	8.22 [5.33, 11.33]				1.05	4.00	0.04	1.78	20.07	81.78	0.68	0.31
dpo	explicit	3.78 [2.22, 5.78] ♦	-4.44 ♦	35 : 15	7e-03	0.36 (1.05)	1.56 (4.00)	0.01 (0.04) ♦	0.49 (1.78) ♦	29.13 (20.07)	81.98 (81.78)	0.59 (0.68)	0.30 (0.31)
SFT (LoRA, merged)	explicit	4.00 [2.22, 6.00]	-4.22	31 : 12	5e-03	0.37 (1.05)	1.56 (4.00)	0.01 (0.04)	0.80 (1.78)	28.36 (20.07)	83.07 (81.78) ♦	0.65 (0.68)	0.29 (0.31)
SFT (LoRA, merged)	explicit	4.00 [2.00, 6.22]	-4.22	28 : 9	3e-03	0.32 (1.05) ♦	1.33 (4.00) ♦	0.01 (0.04)	0.80 (1.78)	18.93 (20.07)	81.58 (81.78)	0.65 (0.68)	0.26 (0.31) ♦
dpo	explicit	6.22 [3.78, 8.89]	-2.00	30 : 21	0.26	0.83 (1.05)	4.89 (4.00)	0.07 (0.04)	1.03 (1.78)	11.08 (20.07)	76.10 (81.78)	0.68 (0.68)	0.43 (0.31)

Values: mitigated run, baseline in brackets, both on exactly the rows the run evaluated. Unstable sets: share with its own 95% CI (bootstrap over candidates); Δ pp: change against the baseline; bold coloured Δ = p < 0.05 (exact sign test on fixed vs broken sets, uncorrected). A run with parse failures is compared on its complete sets only; its own baseline on those sets is shown under its value. Italic ♦ = best value in the column within its condition (real mitigations only; hire rate and FS have no better direction).

Fixed : broken — each counterfactual set (one candidate-job pair, one condition, all attribute variants) paired with the same set at baseline. Fixed: unstable at baseline (the attribute alone changed the decision), stable after the mitigation. Broken: stable at baseline, unstable after — the mitigation introduced a new dependence on the attribute. Δ unstable sets = (broken – fixed) / sets; e.g. 58 : 57 means the method fixes about as many sets as it breaks.

* Lexical scrubbing is an oracle upper bound, not a comparable mitigation: it removes the attribute with the study's own injection templates, so every variant becomes character-for-character the attribute-free CV (100% of rows) and the model sees one identical prompt — zero instability by construction. Real CVs do not use our templates; read this row only as the ceiling. LLM scrubbing, which must find the attribute itself, is the realistic version.